



THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

OBJECT ORIENTED
PROGRAMMING
LAB Assignment 1

SUBMITTED BY:

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QUESTION1: Write a program (WAP) to display "Hello World" on console display.

INPUT:

```
#include <iostream>
using namespace std;
int main(){
    cout<<"Hello World"; return
    0;
}
```

```
Hello World
```

```
=== Code Execution Successful ===
```

QUESTION2: WAP to that reads two values from keyboard and displays the larger value on the screen.

INPUT:

```
#include <iostream>
using namespace std;
int main() {
    int a,b;
    cout<<"enter two numbers ";
    cin>>a>>b;
    if (a>b){
        cout<<"the greater number is "<<a;
    }
    else if (b>a){
        cout<<"the greater number is "<<b;
    }
    else cout<<"both are equal"; return
    0;}

```

```
enter two numbers 23 25
the greater number is 25
```

```
=== Code Execution Successful ===
```

QUESTION3: WAP to read three values a, b, c from keyboard and display the value of $x = a/b - c$.

INPUT:

```
#include <iostream>
using namespace std;
int main() {
    int a,b,c,x;
    cout<<"enter three numbers
"; cin>>a>>b>>c; x = a/b - c;
    cout<<"the value of x: "<<x;
```

```
enter three numbers 1 1 1
the value of x: 0
```

```
=== Code Execution Successful ===
```

QUESTION4: WAP to implement the following control characters.

(\n, \t, \a, \r)

INPUT:

```
#include <iostream>
using namespace std;
```

```
int main() { cout <<
"Hello\nWorld\n";
cout << "Hello\tWorld\n";
cout << "\a"; cout
<< "\rStart"; return
0;
}
```

Your Output

```
Hello
World
Hello World
Start
```

QUESTION 5: Implement namespace in a program to illustrate the use of same name variables and functions in different sections/libraries of the code.

INPUT:

```
#include <iostream> using
namespace std;
```

```
namespace First {
int x =
    10; void
show() {
    cout << "First namespace value = " <<
x<<endl;
}
}
```

```
namespace Second { int x =
    20; void show() { cout <<
"Second namespace value = "
<< x<<endl;
}
}
```

```
int main() {
    cout<<First::x<<endl;

    cout<<Second::x<<endl;
    First::show();
    Second::show();
    return 0; }
```

Your Output

```
10
20
First namespace value = 10
Second namespace value = 20
```

QUESTION6: Write a program to define a structure Student that contains 3 variables and 2 functions: - Variables: (a) name (b) roll (c) marks - Functions: (1) setStudentData() (2) getStudentData() Use loops to input and output data for 3 student object entries.

INPUT:

```
#include <iostream>
using namespace std;

struct Student {
    string name;
    int roll;
    float marks;

    void setStudentData() {
        cout << "Enter name, roll, marks: ";
        cin >> name >> roll >> marks;
    }

    void getStudentData() {
        cout << name << " " << roll << " " << marks
        << endl;
    }
};

int main() {
    Student s[3];

    for (int i = 0; i < 3; i++)
        s[i].setStudentData();

    for (int i = 0; i < 3; i++)
        s[i].getStudentData();

    return 0;
}
```

```
Enter name, roll, marks: HARSHITA 205 89
Enter name, roll, marks: MRIDUL 204 76
Enter name, roll, marks: PRANJAL 208 88
HARSHITA 205 89
MRIDUL 204 76
PRANJAL 208 88
```

```
=== Code Execution Successful ===
```

QUESTION7: In the previous program make the data private, and functions public.

INPUT:

```
#include <iostream>
using namespace std;
```

```
struct Student
```

```
{ private:
```

```
string name;
```

```
int roll;
```

```
float marks;
```

```
public: void setStudentData()
```

```
{
```

```
    cout << "Enter name, roll, marks: ";
```

```
cin >> name >> roll >> marks;
```

```
}
```

```
void getStudentData() {
```

```
    cout << name << " " << roll << " " << marks
```

```
<< endl;
```

```
}
```

```
};
```

```
int main() {
```

```
Student s[3];
```

```
    for (int i = 0; i < 3; i++)
```

```
s[i].setStudentData();
```

```
    for (int i = 0; i < 3; i++)
```

```
s[i].getStudentData();
```

```
    return 0;
```

```
}
```

```
Enter name, roll, marks: HARSHITA 205 23
```

```
Enter name, roll, marks: MRIDUL 204 56
```

```
Enter name, roll, marks: PRANJAL 208 26
```

```
HARSHITA 205 23
```

```
MRIDUL 204 56
```

```
PRANJAL 208 26
```

```
=== Code Execution Successful ===
```

QUESTION8: Convert program #4 from struct to class.

INPUT:

```
#include <iostream>
using namespace std;
```

```
class Student {
```

```
private:
```

```
string name;
```

```

    int roll;
    float marks;

public:
    void
    setStudentData() {
        cout << "Enter name, roll, marks: ";
        cin >> name >> roll >> marks;
    }

    void getStudentData() {
        cout << name << " " << roll << " " << marks <<
endl;
    }
};

int main() {
    Student s[3];

    for (int i = 0; i < 3; i++)
        s[i].setStudentData();

    for (int i = 0; i < 3; i++)
        s[i].getStudentData();

    return 0;
}

```

```

Enter name, roll, marks: HARSHITA 205 23
Enter name, roll, marks: MRIDUL 206 25
Enter name, roll, marks: PRANJAL 205 78
HARSHITA 205 23
MRIDUL 206 25
PRANJAL 205 78

=== Code Execution Successful ===

```

QUESTION9: Discuss the differences of structures in C and structures in C++ during the lab with other students or your instructor.

ANSWER9: Structures in C mainly group related data. They cannot contain functions or support object-oriented features. All members of a C structure are public, so data hiding is not possible. In contrast, structures in C++ can have both data members and member functions. C++ structures support access specifiers like public and private. This enables encapsulation and data security. Overall, structures in C++ are more powerful and are used like classes. This is different from the simpler, data-only structures in C.

QUESTION10: WAP to read an integer value from the keyboard and display on screen “WELL DONE” that many times.

INPUT:

```

#include <iostream>
using namespace std;

```

```

int main() {
    int x,i;
    cout<<"enter a number ";
    cin>>x;
    for(i=1;i<=x;i++){
        cout<<"WELL DONE \n";
    }
    return 0;
}

```

```

enter a number 2
WELL DONE
WELL DONE

```

=== Code Execution Successful ===

QUESTION11: WAP to utilize assignment += and -= type operators.

INPUT:

```

#include <iostream>
using namespace std;

```

```

int main() {
    int a = 10;
    a += 5;
    cout << "After += operator, a = " << a <<
endl;
    int b = 8;
    b -= 3;
    cout << "After -= operator, b = " << b <<
endl;

    return 0;
}

```

```

After += operator, a = 15
After -= operator, b = 5

```

=== Code Execution Successful ===

QUESTION12: WAP to swap two numbers using bitwise operator.

INPUT:

```

#include <iostream>
using namespace std;

```

```

int main() {
    int a, b;
    a = 10;  b
= 11;  a =
a ^ b;  b =
a ^ b;
    a = a ^ b;

    cout << "After swapping:" << endl;
    cout << "a = " << a << ", b = " << b;

```

Your Output

```

After swapping:
a = 11, b = 10

```


QUESTION13: Write a program to solve the following problem:

A library charges a fine for every book return late. For first 5 days the fine is 50 paise, for 6–10 days fine is one rupee and above 10 days fine is 5 rupees. If you return the book after 30 days your membership will be cancelled. WAP to accept number of days the member is late to return the book and display the fine or appropriate message.

INPUT:

```
#include <iostream>
using namespace std;
```

```
int main() {
    int days;
    cout << "Enter number of days late: ";
    cin >> days;
```

```
    if (days <= 5)    cout <<
    "Fine: 50 paise";  else if
    (days>=6 && days <= 10)
    cout << "Fine: 1 rupee";  else if
    (days>10 && days <= 30)
    cout << "Fine: 5 rupees";  else
    cout << "Membership cancelled";
```

```
    return 0;
}
```

```
Enter number of days late: 11
Fine: 5 rupees
```

```
=== Code Execution Successful ===
```